Summary

1 **Atom embeddings.** MACE atom embeddings are element-resolved (10-NN element agreement 92% for the MH1 readout, ≈100% for every interaction layer); UMA embeddings are organised by local environment (η^2(element) = 0.25).

2 **Atom matching.** Between a cell and its supercell, fused Gromov–Wasserstein (FGW) with a periodic neighbour kernel retains 69% of neighbour pairs, against 52% for embedding-only optimal transport (OT) and a ceiling of 74%. The embedding used has no significant effect.

3 **Structure distance.** A size-intensive FGW distance retrieves the same material across 2× and 4× supercells in 97% of cases.

4 **MOF-off vs OMat24.** The datasets share elements but not chemistry: 2 of 2,552 MOF-off compositions occur in OMat24, and 83–93% of MOF-off atoms lie in embedding regions that OMat24 samples at least ten times more sparsely.

5 **Analogue search.** Among 10M OMat24 structures, the closest analogue of every MOF query is a near-equilibrium AIMD frame; rattled 300/500/1000 frames distort C–H bonds (62–64% deviate by more than 0.15 Å).

Four embedding representations cover 101M OMat24 and 13.5M external structures

Representation	Model	Layer	Width	Per atom
UMA-128	uma-s-1p2	Input to the energy block	128	Yes
OMAT0-128	mace-omat-0-medium	Last-layer invariants	128	No
MH1-96	mace-mh-1	Readout hidden units, 6 heads	96	Yes
MH1-1024	mace-mh-1	Invariants of both interaction layers	1,024	No

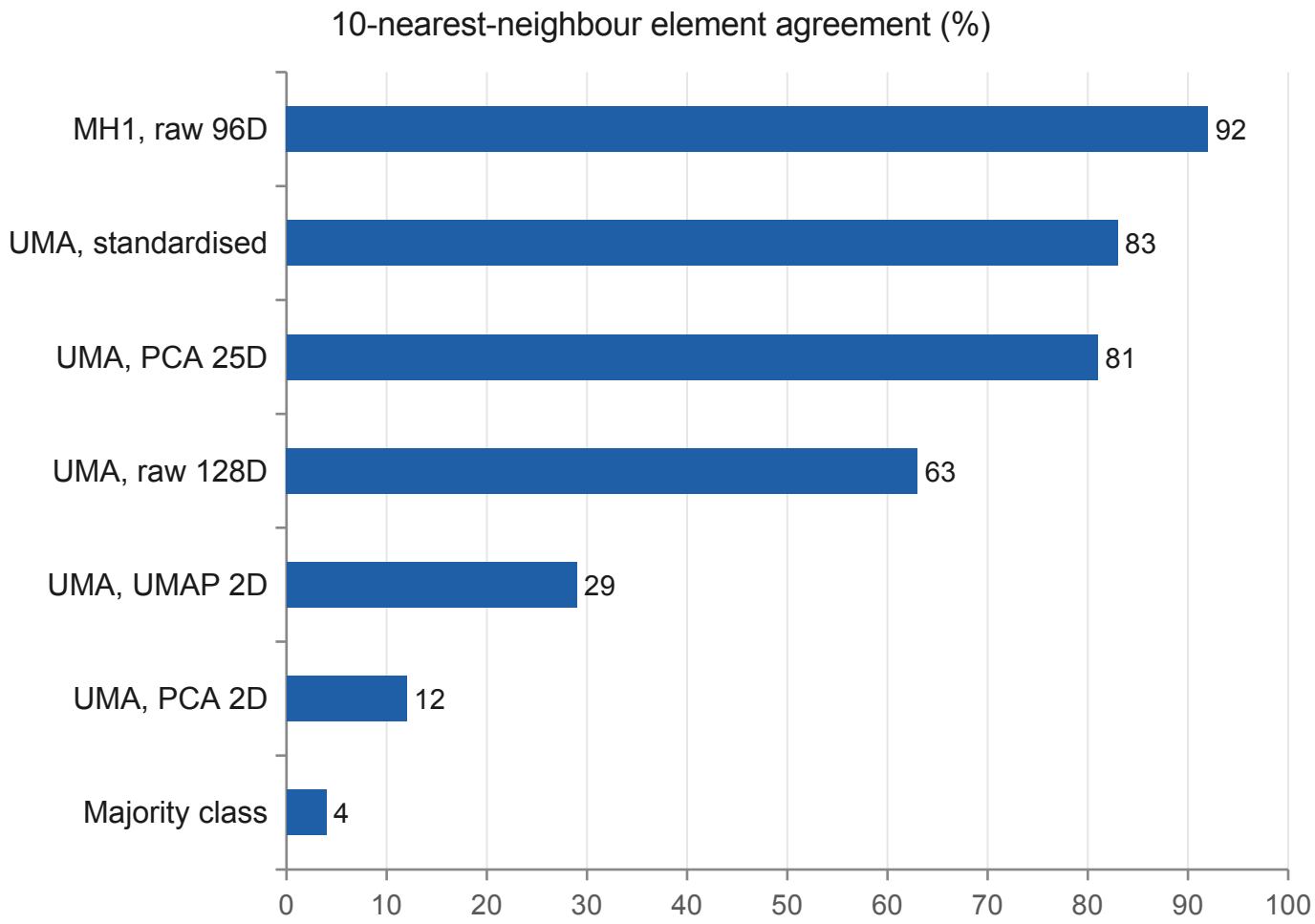
Dataset	Structures	Per-atom sample
OMat24	100.8M	1M and 10M
AM	12.0M	10%
MP-ALOE	910k	10%
MatPES	388k	10%
MAD	96k	10%
MOF-off (r^2SCAN)	80.6k	10%

Extensive structure vectors: $H = \Sigma_i Z_i h_i$ grows with the number of atoms in the cell.

Coordinates: read from the source LMDB and trajectory files; bank row i corresponds to frame i.

Exclusions: 468 group-restricted files (40.6 GB); 12,275 sr_10m frames with interatomic distances below 0.8 Å.

MACE-MH1 atom embeddings are element-resolved; UMA embeddings encode local environment



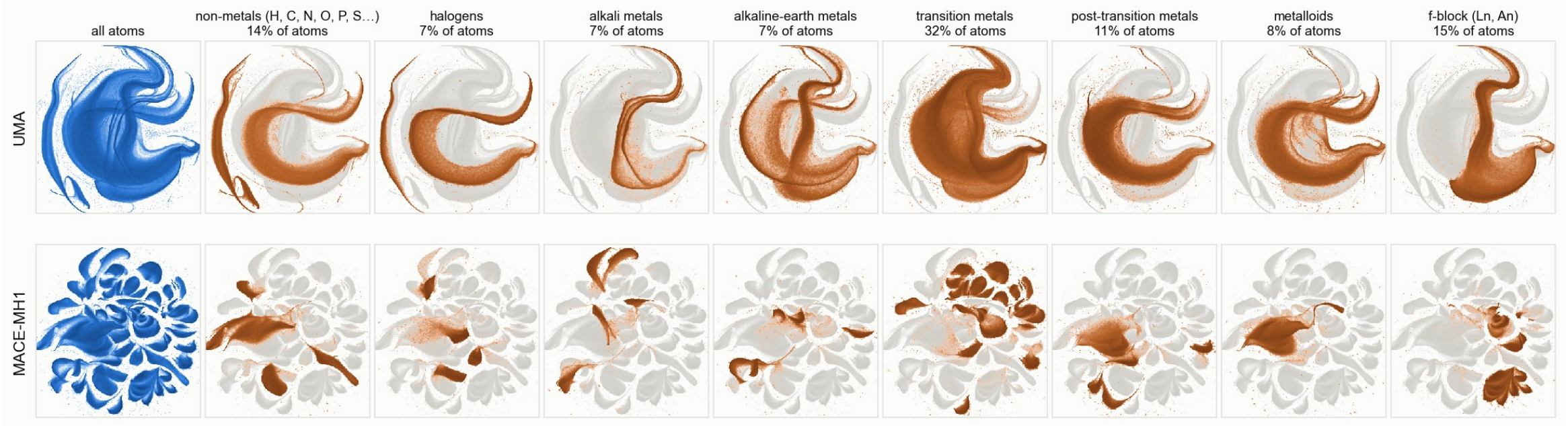
η^2(element): 0.79 for MH1, 0.26 for UMA.

UMA scaling: a few high-variance channels track the OMat24 subset; standardisation raises agreement from 63% to 83%.

No clustering by structure: only 0.3% of neighbours are other-element atoms of the same structure.

93,633 atoms, 5,000 OMat24 sr_1m structures.

In two dimensions, MH1 resolves element islands whereas UMA forms one continuous manifold



86%

2D element agreement, MH1

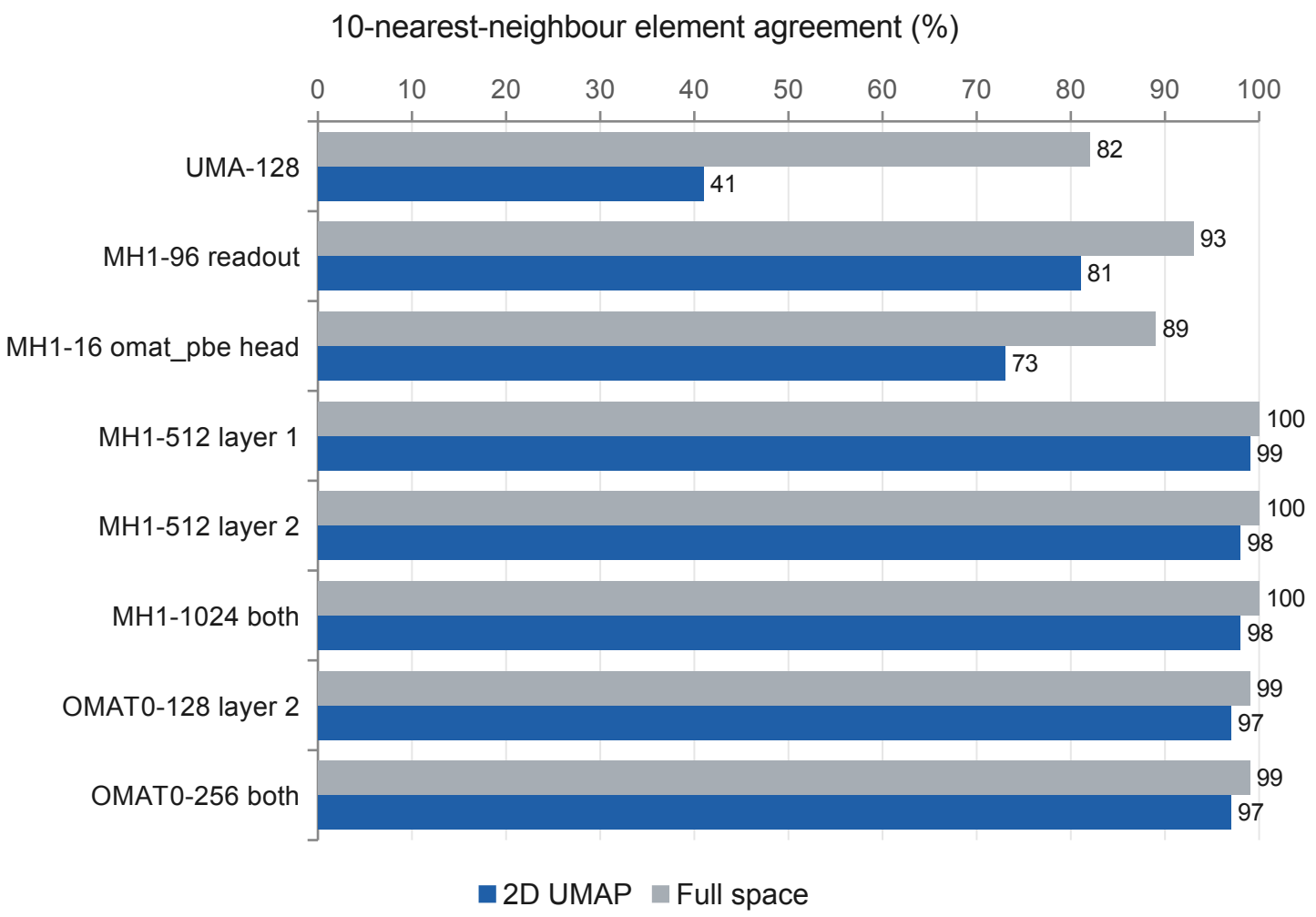
29–30%

2D element agreement, UMA (joint or OMat24-only fit)

18.7M

atoms per map; cuML UMAP, k = 15, min_dist = 0.1

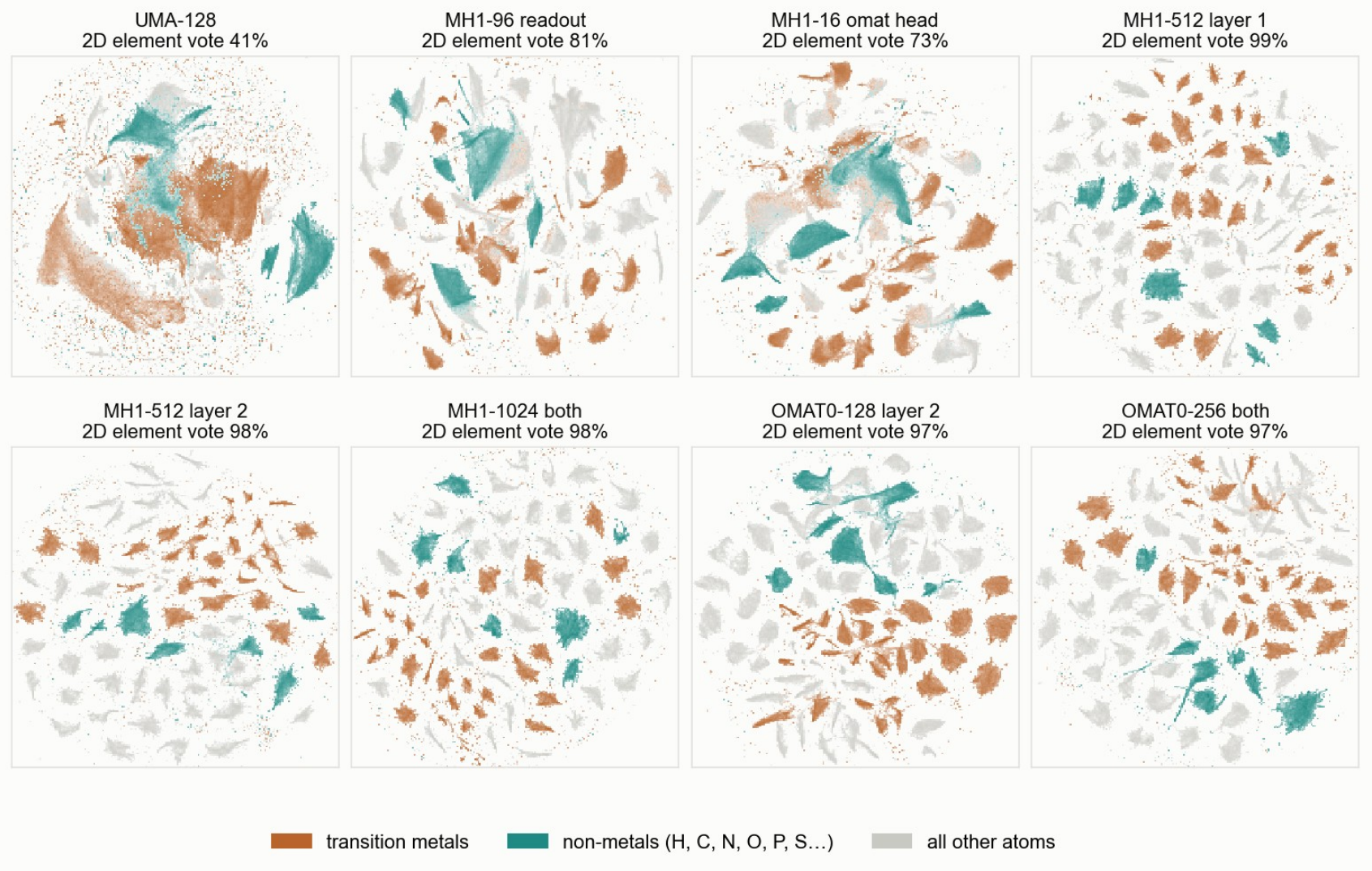
Every MACE interaction layer is effectively an element label



Variant	η²(element)
UMA-128	0.25
MH1-96 readout	0.41
MH1-16 omat_pbe head	0.40
MH1-512 layer 1	0.81
MH1-512 layer 2	0.68
OMAT0-128 layer 2	0.53

50,188 OMat24 sr_1m structures (927,070 atoms); features standardised per channel. The extracted descriptors reproduce the archived per-structure vectors to 3×10^{-8}.

Only UMA retains a connected representation in two dimensions

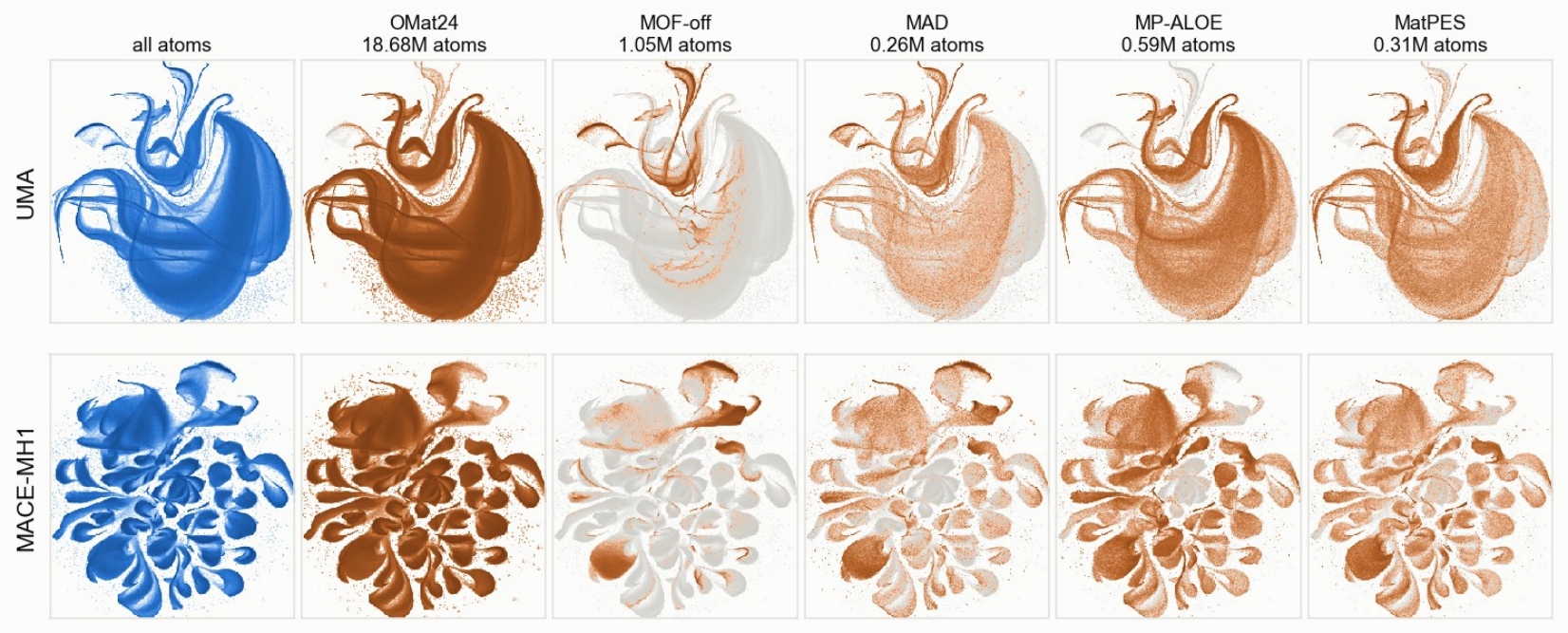


UMA: families form broad, connected regions ordered by environment.

MH1 readout: element islands plus regions shared by similar environments.

Interaction layers: one compact island per element; environment variation is confined within islands.

MOF-off atoms occupy regions of embedding space that OMat24 samples sparsely

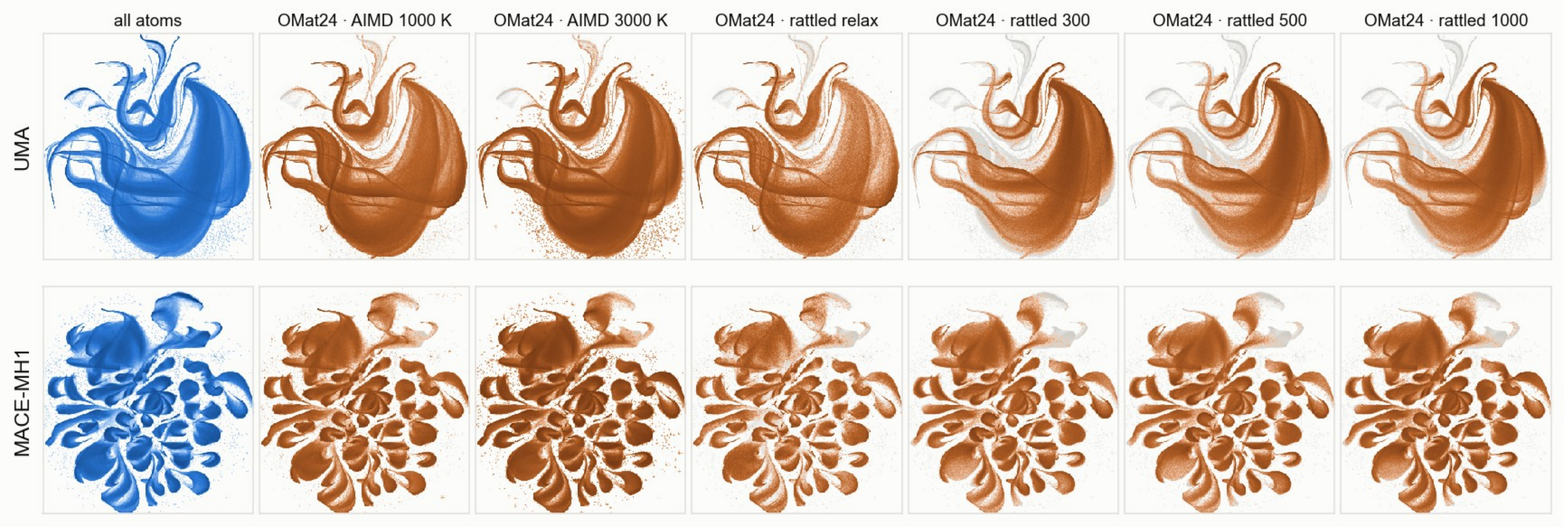


Dataset	UMA	MH1
MOF-off	93%	83%
MAD	42%	43%
MatPES	11%	14%
MP-ALOE	7%	10%

Fraction of each dataset's atoms in map pixels where it is ≥ 10× enriched relative to OMat24.

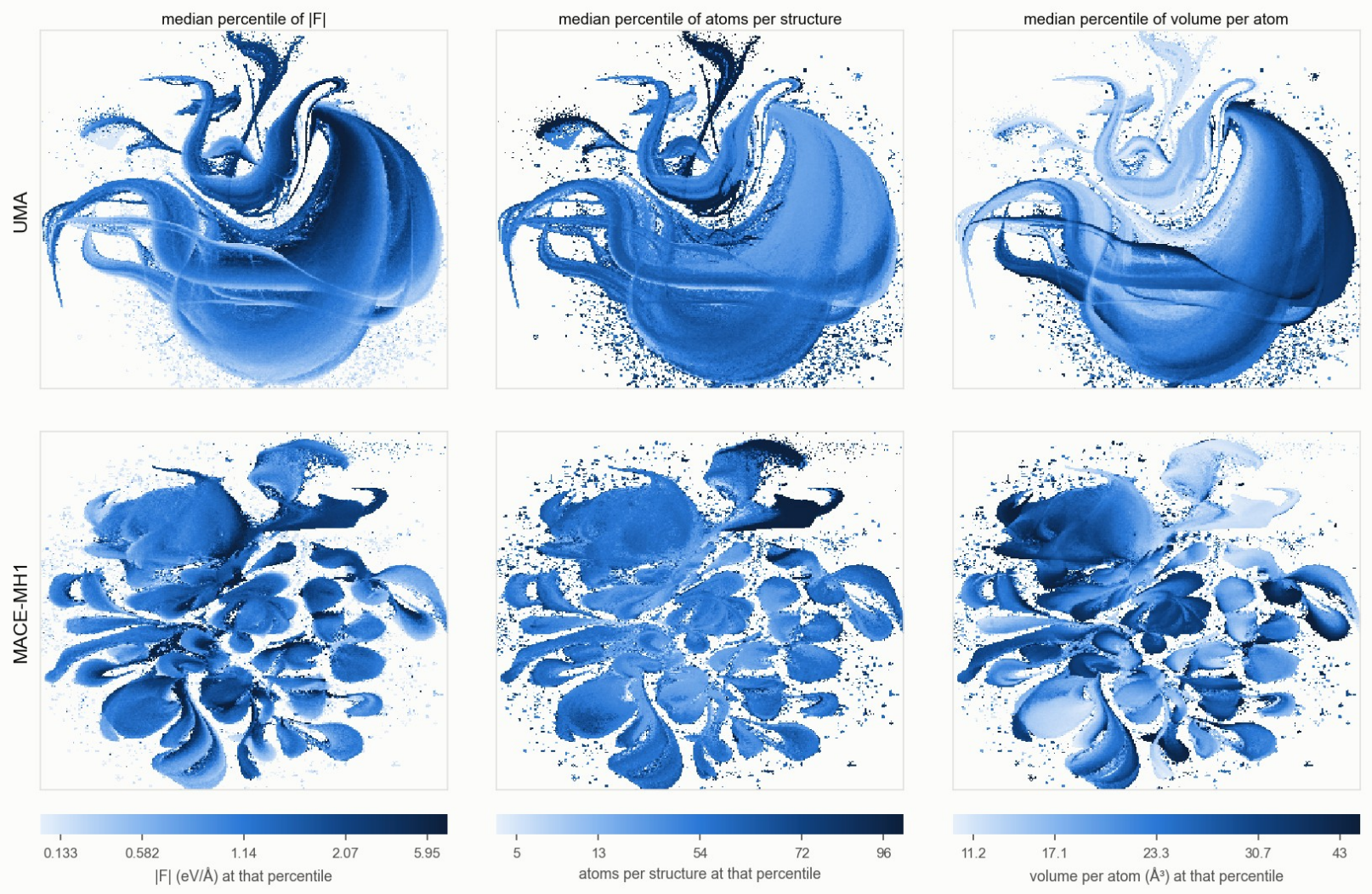
Joint fit over 20.9M atoms: OMat24 sr_1m and 10% samples of MOF-off, MAD, MP-ALOE and MatPES. Rows: UMA, MACE-MH1.

The six OMat24 generation protocols populate nearly the same regions



Each column shows one protocol over all atoms (grey). Rows: UMA, MACE-MH1. Rattled 300/500/1000 frames leave regions unpopulated that AIMD and rattled-relax frames occupy.

UMA embeddings vary smoothly with force magnitude and atomic volume



Force: UMA shows a gradient across the main manifold (5th–95th percentile 0.13–6.0 eV/Å); in MH1 it varies within islands.

Cell size: the largest cells, mainly MOF-off (median 170 atoms), form distinct regions in both maps.

Volume per atom: the strongest UMA gradient, separating dense from open frameworks.

Median |F|: 1.1 eV/Å in OMat24, 1.8 eV/Å in MOF-off, 0.5–0.6 eV/Å in MAD, MP-ALOE and MatPES.

A benchmark of 94 materials, each as a primitive cell and a 2× or 4× rattled supercell

1,212
materials with both a relaxed cell and a rattled-300 supercell

10 + 84
development and held-out pairs with a verified alignment

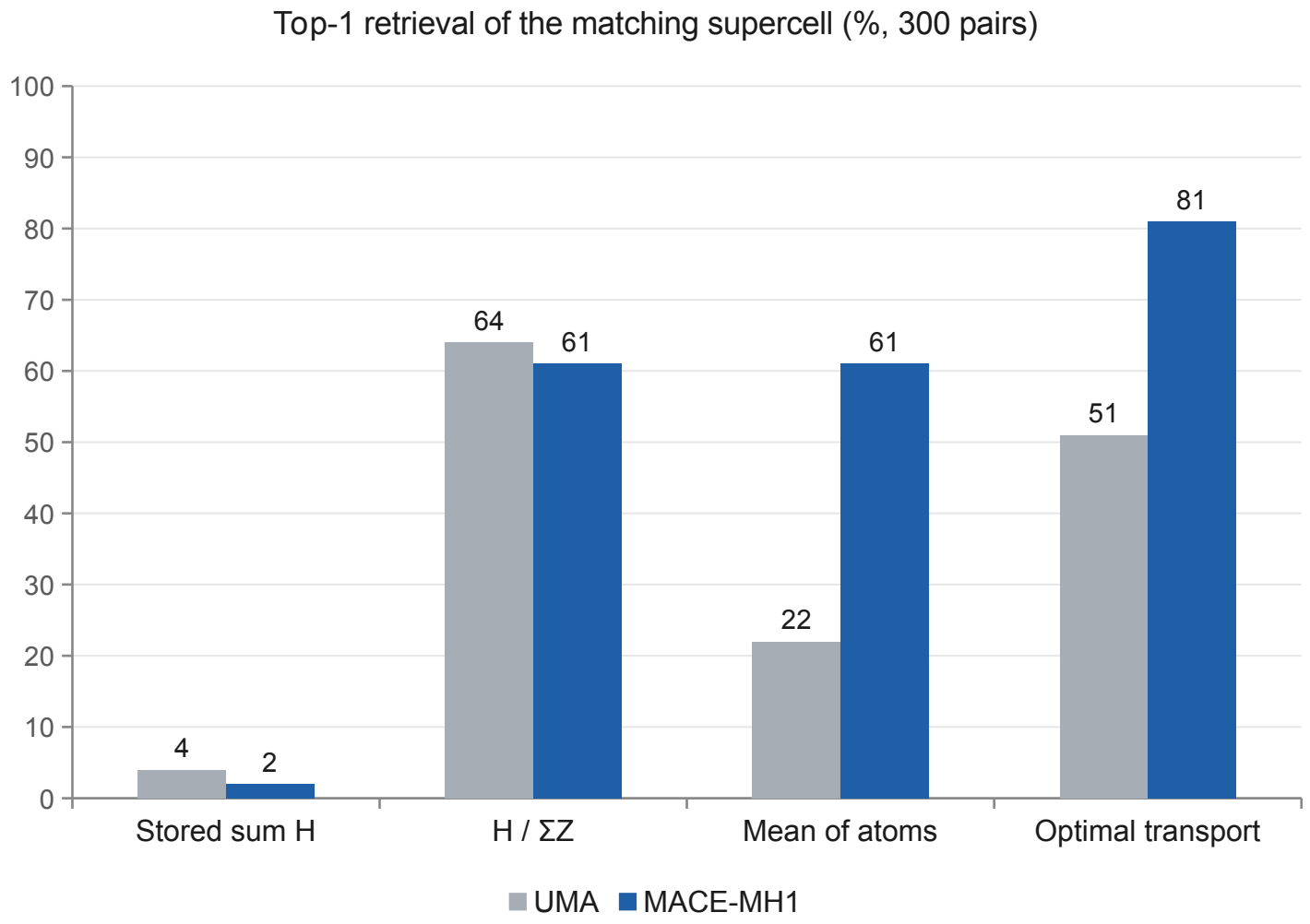
0.6 Å
median displacement of supercell atoms from ideal sites

Pairs: rattled-relax cell (2–8 atoms) and rattled-300 supercell from the same Alexandria parent, with volume per atom within 5%.

Ground truth: pymatgen StructureMatcher alignment of the supercell onto the small cell; 76 of 160 held-out candidates could not be aligned and were excluded.

Protocol: parameters were fixed on the 10 development pairs and evaluated once on the 84 held-out pairs.

Extensive sums fail across cell sizes; normalisation or optimal transport restores invariance



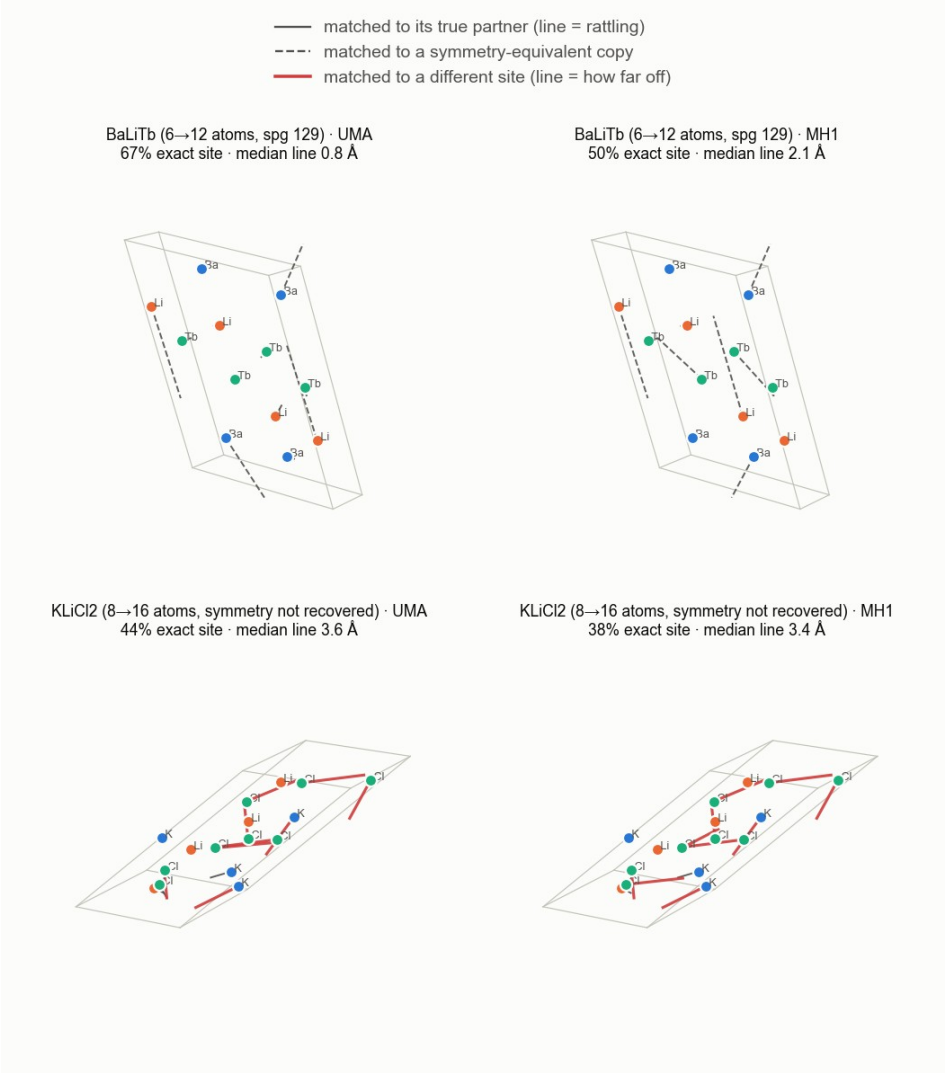
Stored sums: extensive in N; 2–4% top-1.

H / ΣZ: best for UMA (64%).

Optimal transport: atom sets with weights 1/N; best for MH1 (81%).

All 1,212 pairs: 57% (UMA) and 50% (MH1) with H / ΣZ, owing to additional strain.

Embedding-only OT assigns elements correctly but does not preserve local connectivity



98–100%

atoms assigned to the correct element

62% / 53%

exact site, UMA / MH1 (random within element: 49%)

50–54%

neighbour pairs preserved (random within element: 51%)

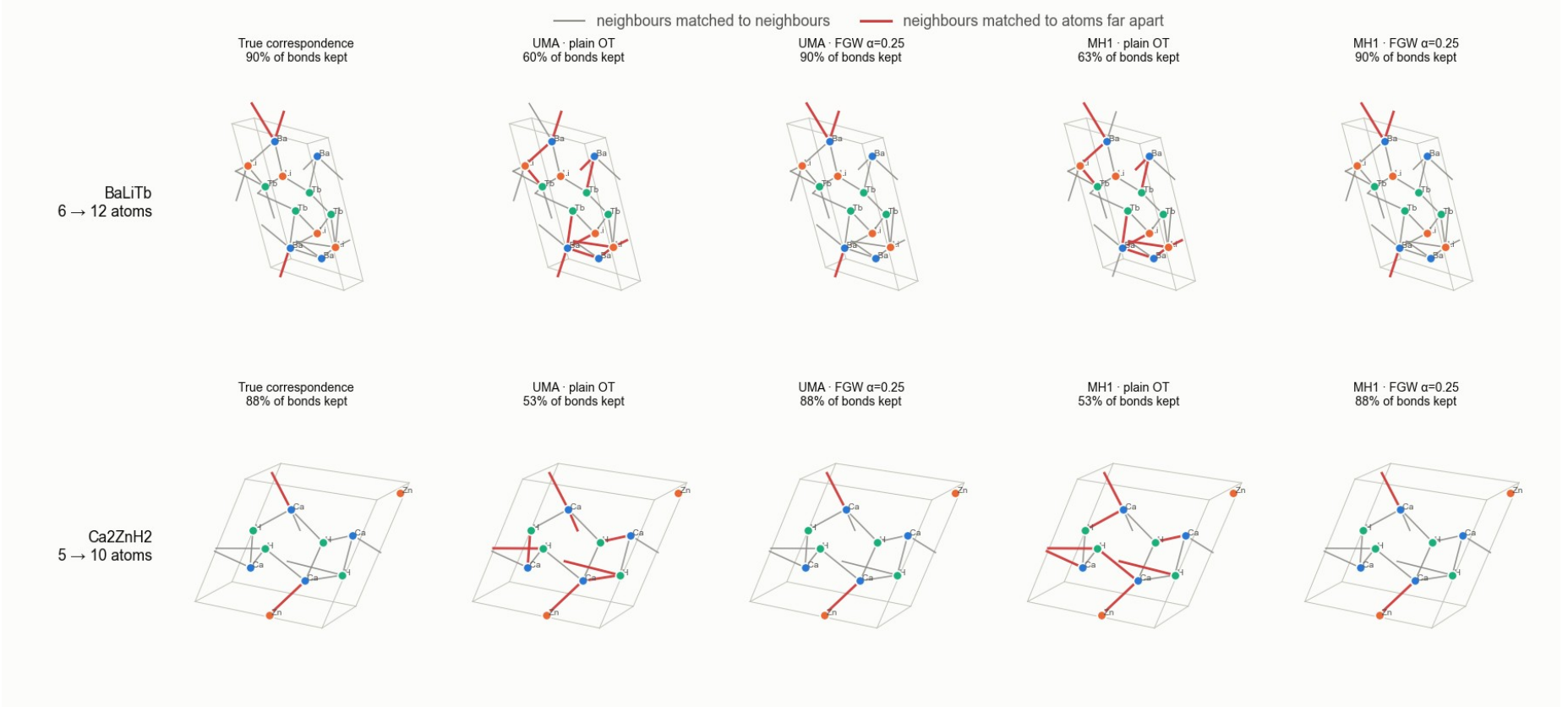
BaLiTb: every error is a symmetry-equivalent site.

$KLiCl_2$: inequivalent K and Cl sites are exchanged over 3–4 Å.

Fused Gromov–Wasserstein adds a periodic neighbour-kernel term to the transport cost

$$\min_T \; (1-\alpha)\,\Sigma\, M_{ij}\, T_{ij} \;+\; \alpha\,\Sigma\,(C_1[i,k]-C_2[j,l])^2\, T_{ij}\, T_{kl}$$

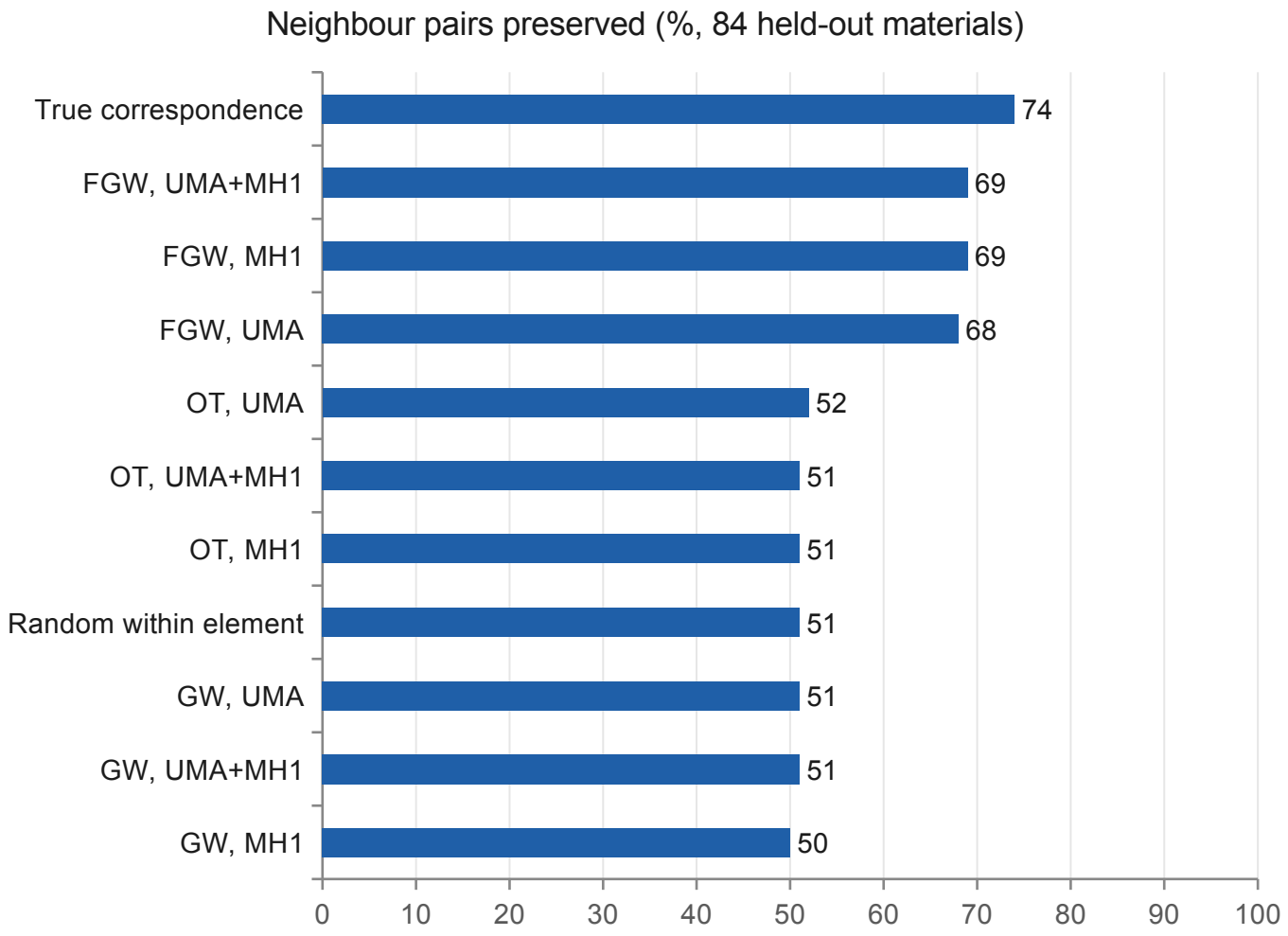
M: squared embedding distance between atoms i and j. C: Gaussian neighbour kernel exp(−d²/2σ²) summed over all periodic images. T: transport plan.



Three modifications were required to apply FGW to periodic crystals

Modification	Problem addressed	Effect
Tile the small cell with a lattice-only integer supercell matrix	Kernels of a cell and its supercell are not comparable	True correspondence preferred by the objective in 10/10 pairs (previously 3/10)
Sum the neighbour kernel over all periodic images	Minimum-image distances retained 23–66% of each neighbourhood	MH1 exact-site rate 45% → 55%
100 random same-element initialisations	Non-convex objective; two starts remained at the OT solution	Neighbour preservation close to the true correspondence

On 84 held-out materials, FGW preserves 69% of neighbour pairs against 52% for OT



Significant: FGW vs OT, +16–18 points (better for 75–79 of 84 materials).

Significant: embedding-only GW is worse than OT by 12–15 points in exact-site rate.

Not significant: α = 0.1 vs 0.25; UMA vs MH1 vs UMA+MH1.

Supercell size: 4× pairs are easier (81–86% exact site) than 2× pairs (57–66%).

Bootstrap CIs and Holm-corrected Wilcoxon signed-rank tests, per material.

Matching accuracy is insensitive to the choice of embedding layer

Variant	Width	Correct element, OT / FGW	Exact site, OT / FGW	Neighbours preserved, OT / FGW
UMA-128	128	96.9 / 97.6	67.3 / 69.9	52.3 / 67.2
MH1-96 readout	96	98.7 / 98.9	68.7 / 70.6	52.1 / 68.3
MH1-16 omat_pbe head	16	97.1 / 98.4	66.7 / 67.9	51.1 / 68.1
MH1-512 layer 1	512	100 / 100	69.0 / 70.6	52.6 / 68.4
MH1-512 layer 2	512	100 / 100	70.6 / 73.8	50.4 / 68.5
MH1-1024 both	1,024	100 / 100	69.9 / 71.5	53.6 / 69.0
OMAT0-128 layer 2	128	99.5 / 99.7	70.6 / 70.5	51.8 / 68.4
OMAT0-256 both	256	99.7 / 100	68.4 / 70.5	51.6 / 68.0

Percentages over 94 materials (10 development, 84 held-out). FGW: periodic kernel, $\alpha = 0.25$, 30 initialisations. No difference from the MH1 readout is significant after Holm correction.

Embedding-only Gromov–Wasserstein fails because no term anchors element identity

Observation	Evidence
Embedding-only GW exchanges entire elements	BaLiTb across models: 0% correct element; the GW objective prefers an incorrect assignment in 49 of 50 fits
UMA + MH1 concatenation gives no measurable gain	Equal to the better single model, never significantly better (blocks scaled by 1/√width)
Symmetry is lost in the rattled data	All rattled-300 supercells are P1 at 0.5 Å tolerance; 15–19% of small cells retain the parent space group
The method is inexpensive	Per pair on one core: OT 0.1 ms; one FGW solve 0.3 ms; 102 initialisations ≈ 30 ms plus 40–60 ms setup

MOF-off and OMat24 share elements but not chemistry

2 of 2,552

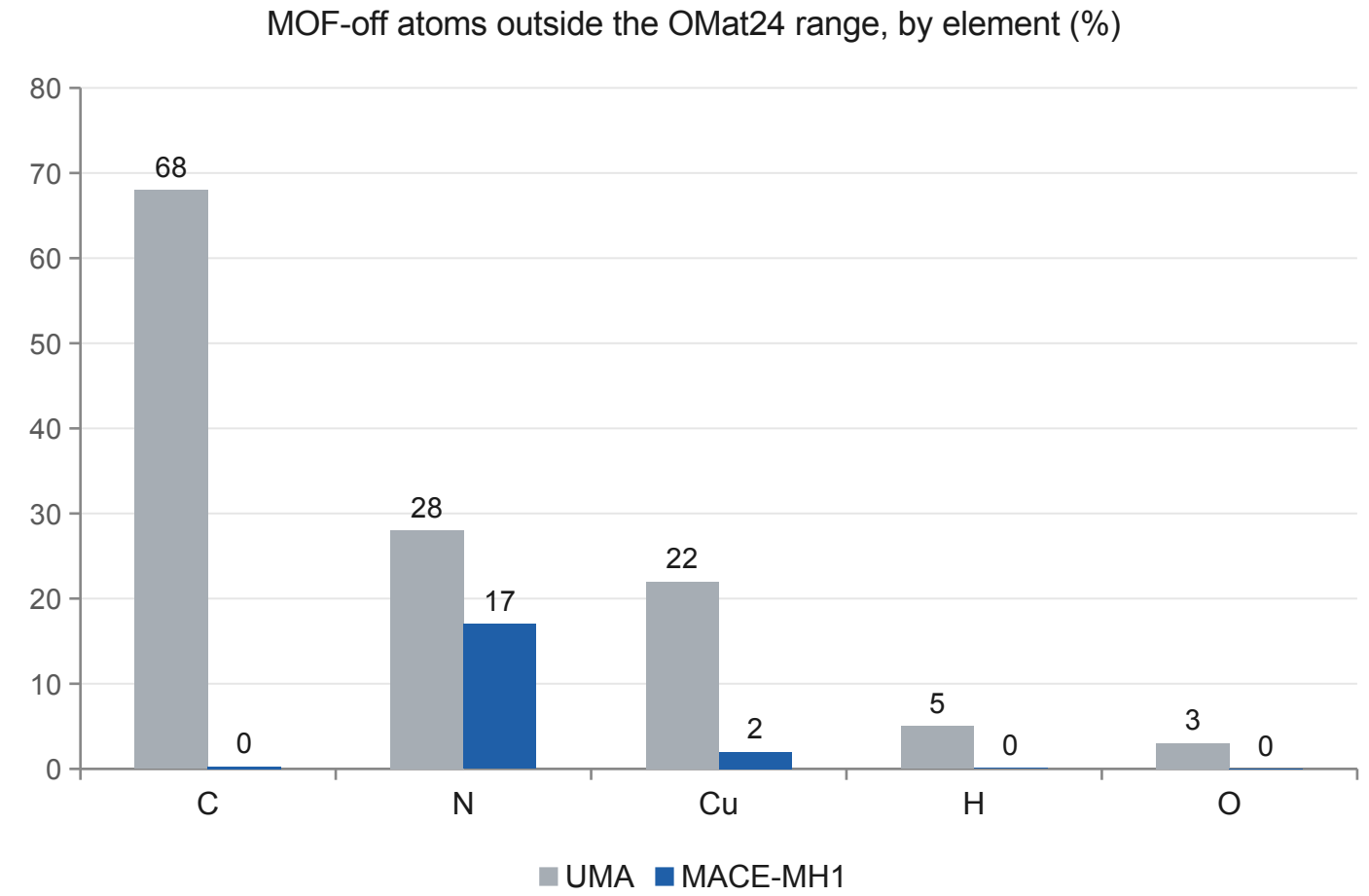
MOF-off compositions present in 10M OMat24 structures

99.4% vs 0.11%

structures containing both C and H (MOF-off vs OMat24)

116 vs 12

median atoms per structure



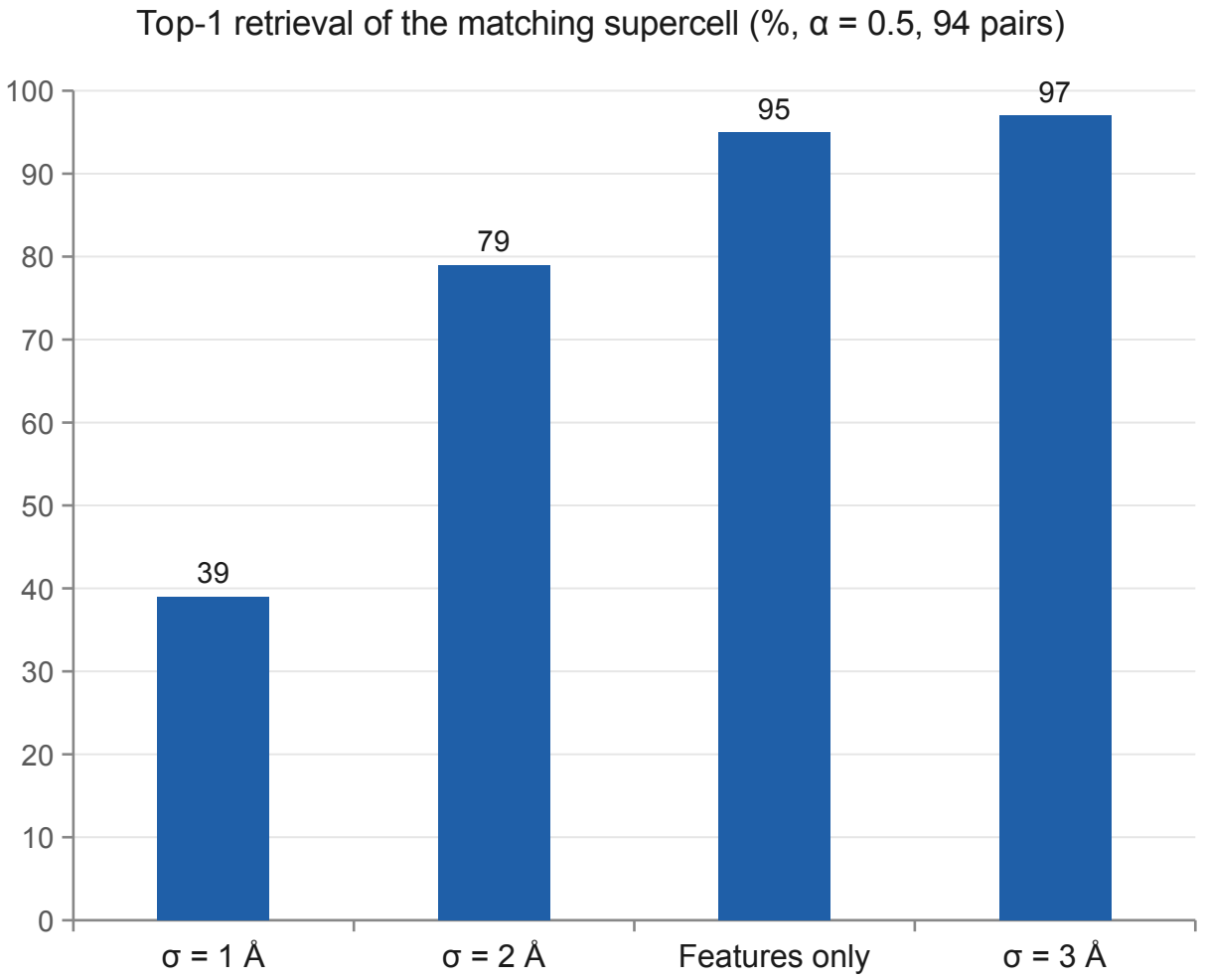
Outside: nearest OMat24 atom farther than the 99th percentile of OMat24 nearest-neighbour distances (20,000 atoms per set).

A size-intensive FGW distance identifies the same material across cell sizes

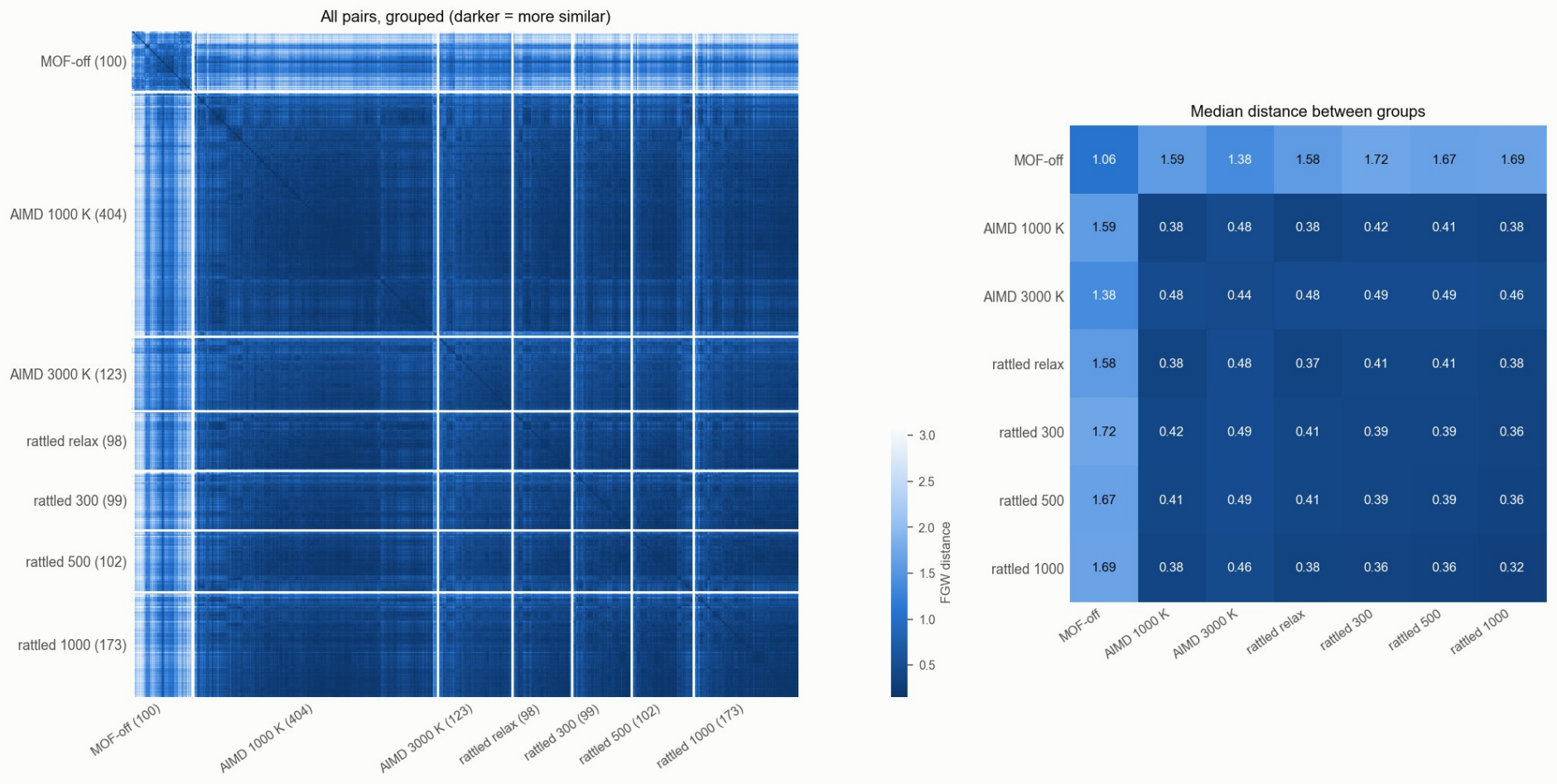
Intensive kernel: $K = N \cdot \Sigma$ over images $\exp(-d^2/2\sigma^2)$, normalised by a single global constant k_0.

Global scaling: feature (m_0) and kernel (k_0) scales are fixed once, so all distances in a matrix are comparable.

Validation: each of 94 small cells must retrieve its own 2× or 4× supercell.

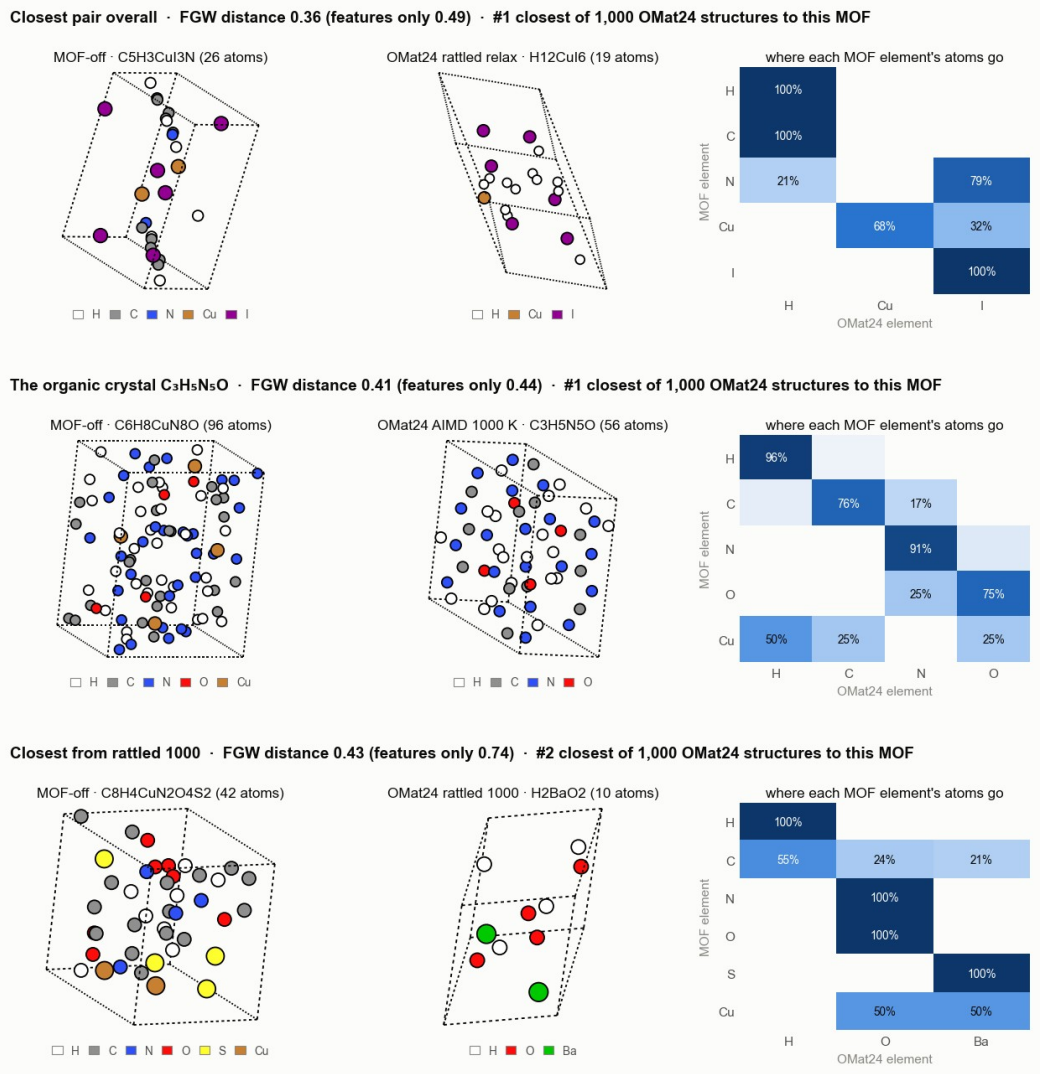


MOF-off forms a distinct block in the pairwise structure-distance matrix



Median distances: MOF–MOF 1.06; OMat24–OMat24 0.41; MOF–OMat24 1.60. 98 of 100 MOFs are closer to another MOF than to any OMat24 structure.

The closest MOF–OMat24 pairs share metal centres, halides or organic fragments



Cu–I framework and $H_{12}CuI_6$: closest pair overall; I → I 100%, Cu → Cu 68%.

Organic crystal $C_3H_5N_5O$: the only C- and H-containing structure in the 1,000-structure sample; nearest neighbour of 80 of 100 MOFs.

$Ba(OH)_2$: N and O are assigned to O, S to Ba; absent elements map to the most similar available element.

A two-stage search ranks 10M OMat24 structures for each of 10 MOF queries

10 MOF queries
10 distinct metals; 42–276 atoms

→ **Surrogate prefilter**
feature OT + 1D W_2 on neighbour densities; ρ = 0.90 with FGW

→ **Top 50,000 per query**
0.5% of the sample; 168,736 distinct structures

→ **Exact FGW re-ranking**
500,000 pairs; UMA features, σ = 3 Å, α = 0.5

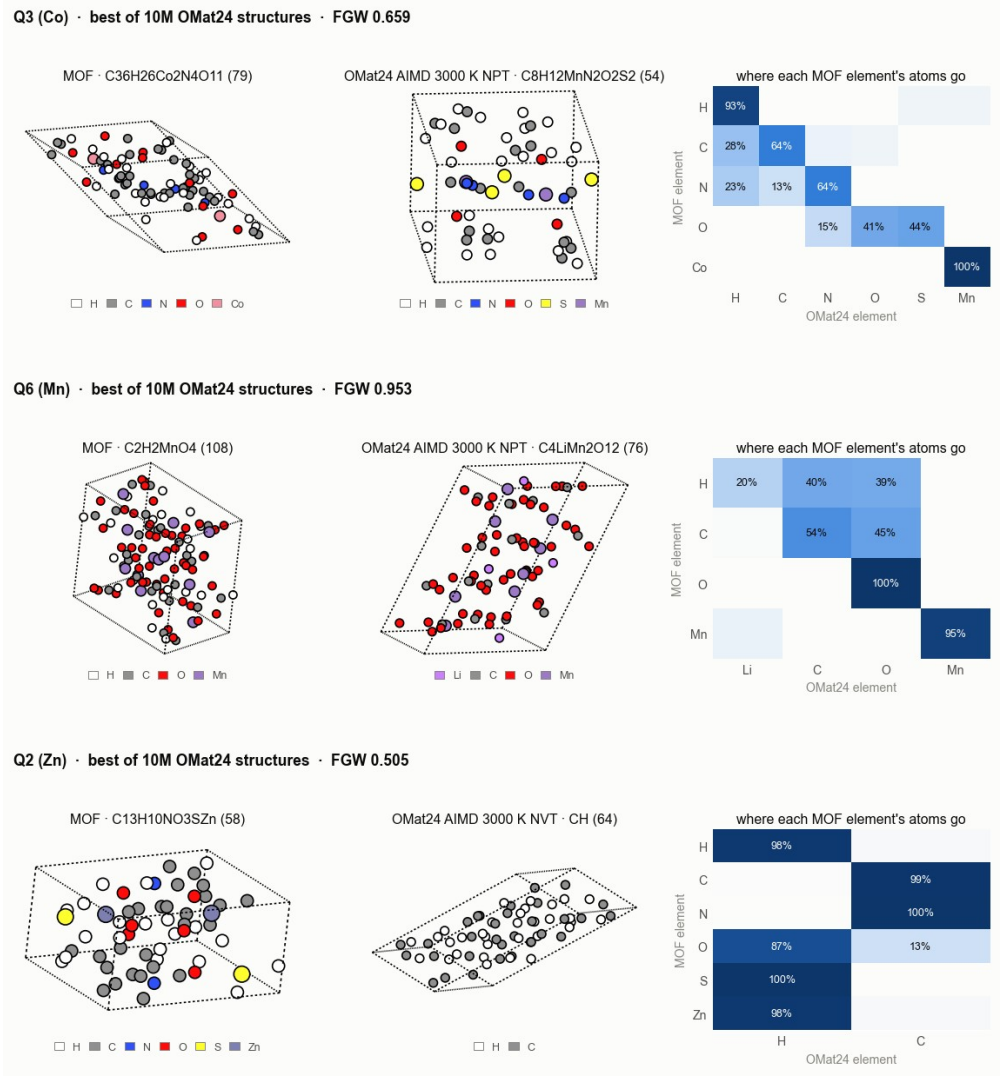
46
core-hours for the prefilter (400 Slurm tasks)

12,275
unphysical frames excluded (atoms closer than 0.8 Å)

≤ 520
prefilter rank of every FGW top-20 match, for 7 of 10 queries

For the three largest MOFs (202–276 atoms), some top-20 matches originated from prefilter ranks of 11,000–35,000; their best matches are poor in any case.

Chemically consistent analogues are recovered for about half of the queries



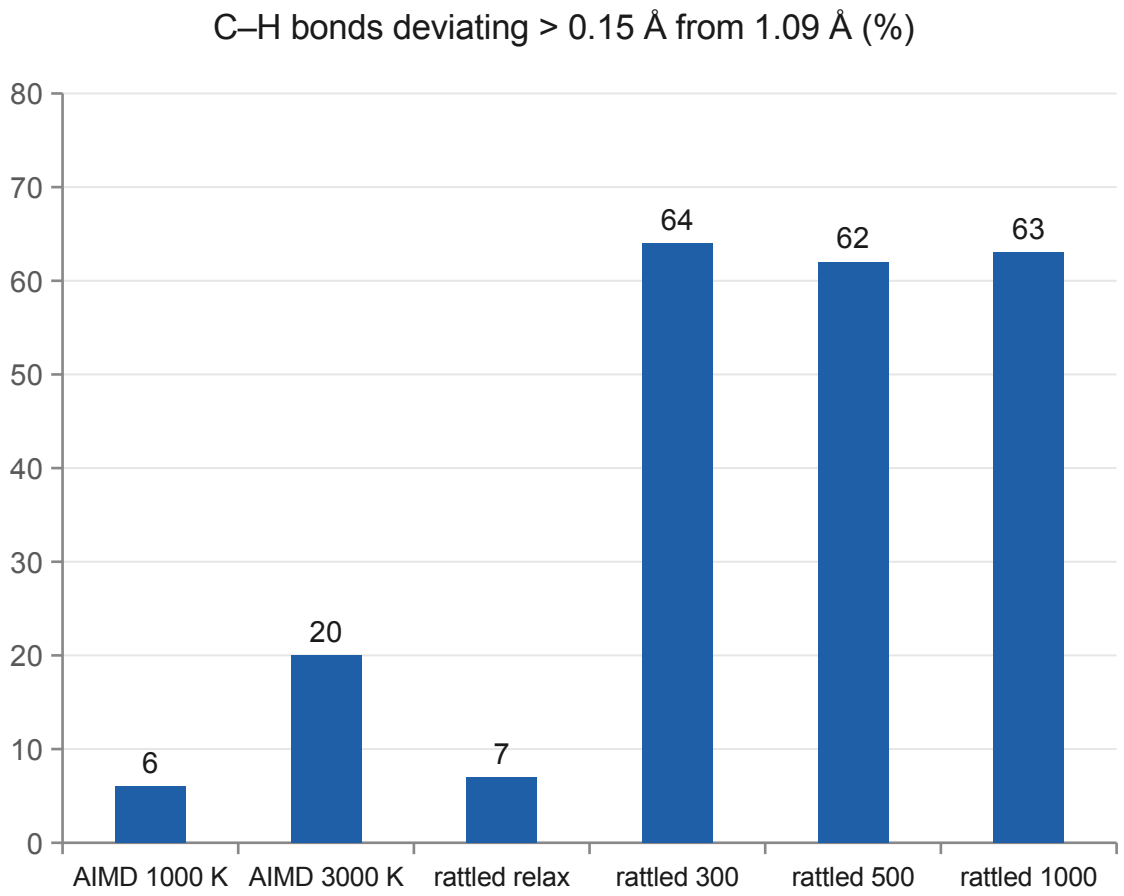
Co MOF and Mn complex: Co → Mn 100%; ligand C, N and O on C, N and O/S.

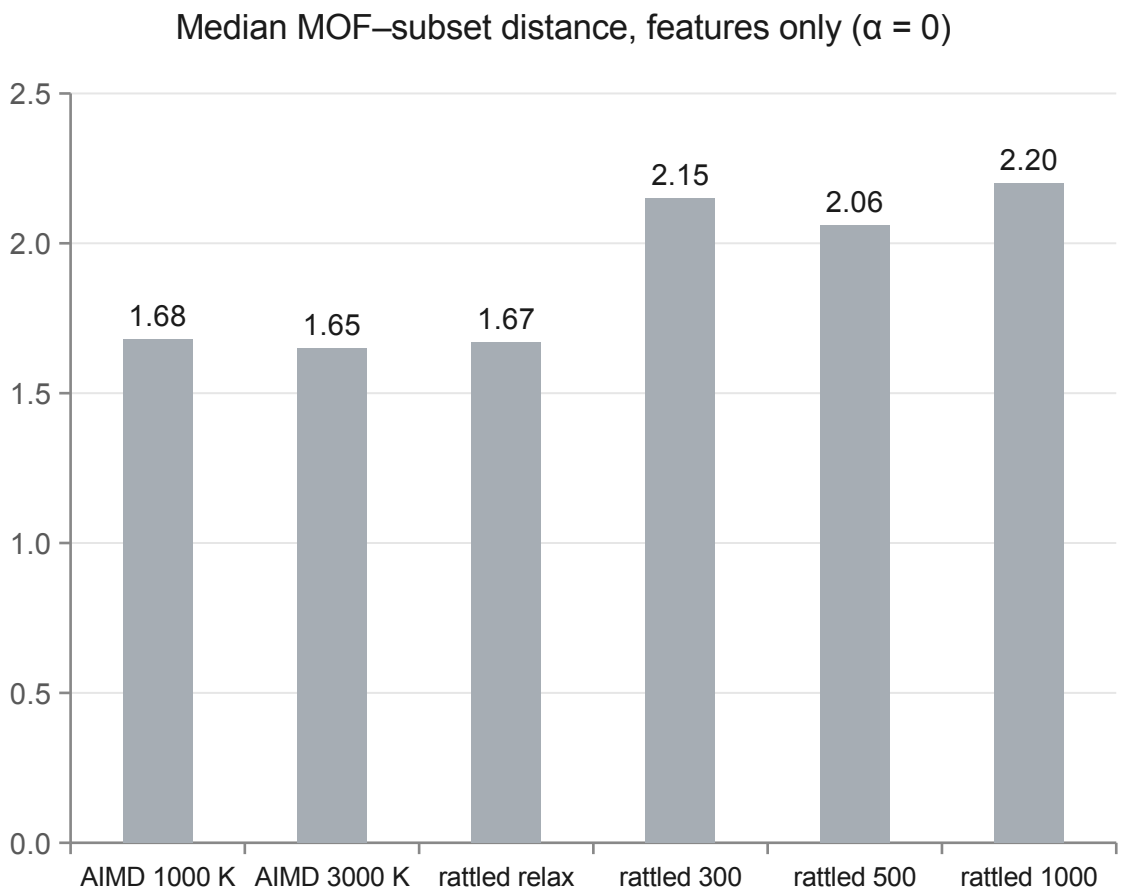
Mn formate and Mn oxalate: Mn → Mn 95%, O → O 100%.

Zn MOF and a 3000 K hydrocarbon: metal, N and O assigned to H or C; uniform atom weights allow C and H to outweigh the metal.

No counterpart: the Ag, Ni and Ce MOFs (202–276 atoms) match only large organosilicon or organometallic cells.

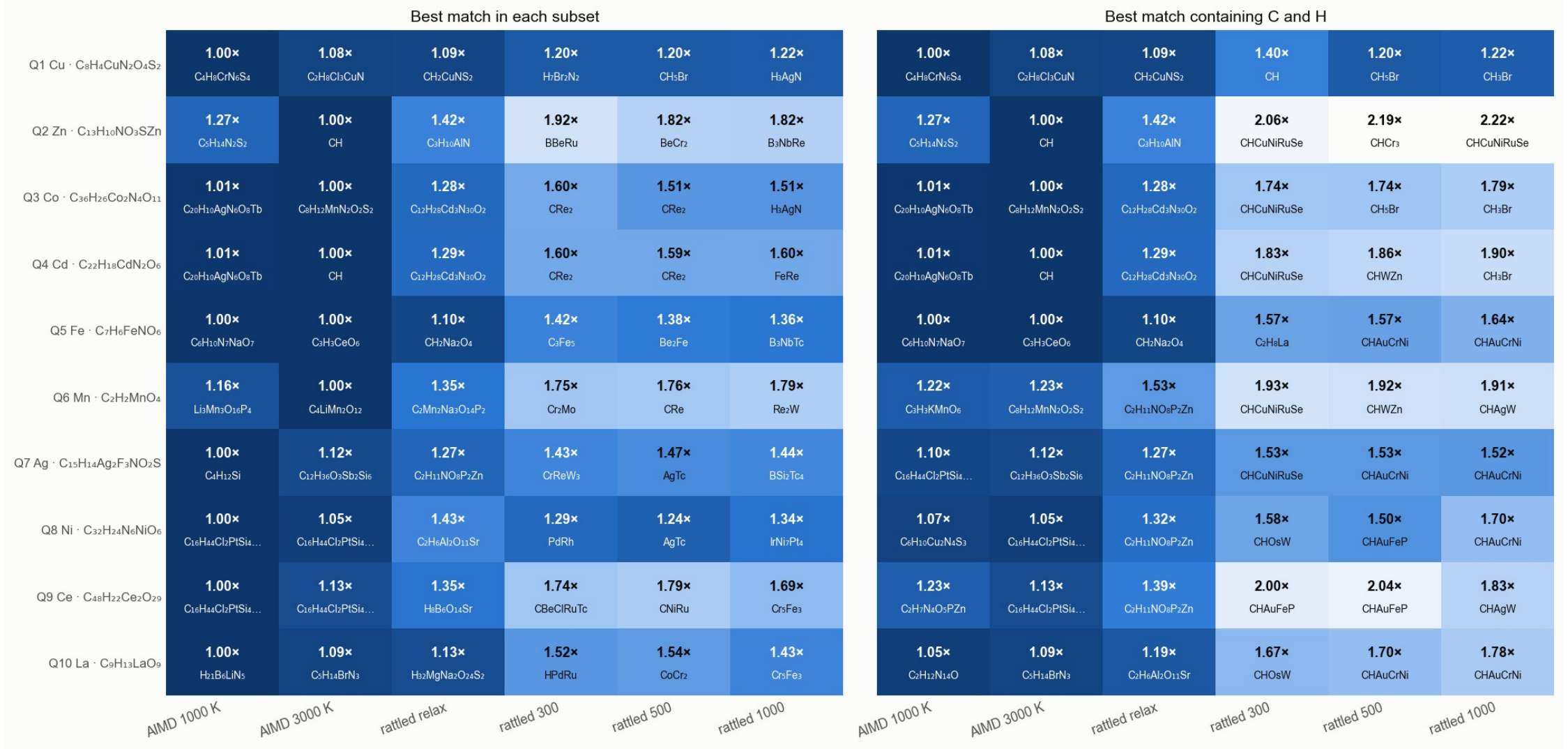
Rattling distorts C–H bonds; near-equilibrium frames match MOFs best



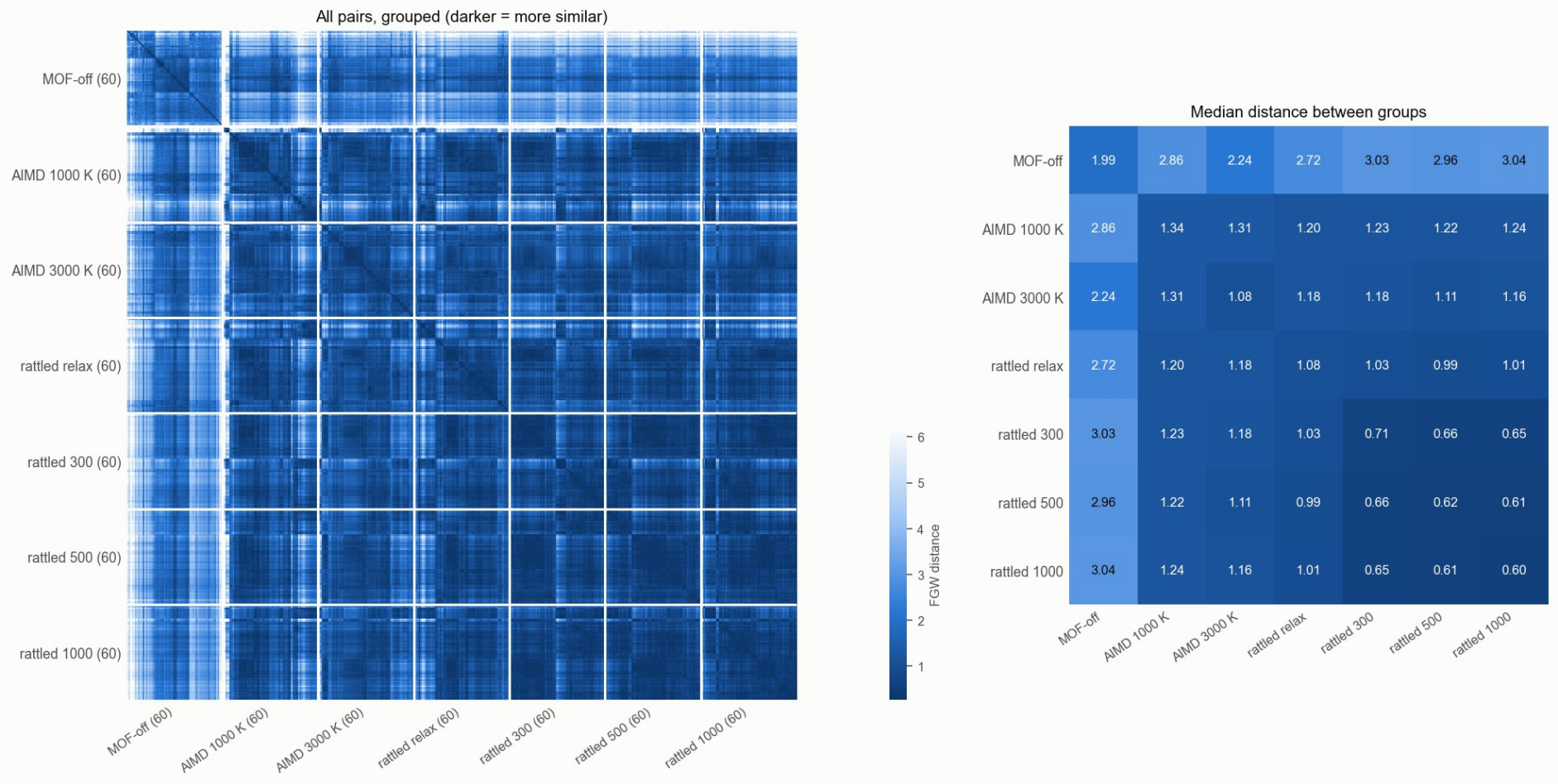


Both panels: OMat24 structures containing C and H only. Distances: 60 structures per protocol.

For every query, the closest match lies in an AIMD subset



With composition controlled, rattled organic frames cluster together and away from MOF-off



Features only (α = 0): MOF–AIMD 1000 K, 3000 K and rattled relax 1.65–1.68; MOF–rattled 300/500/1000 2.06–2.20; MOF–MOF 0.57.

Conclusions and next steps

Conclusions

- Geometry is required for atom correspondence: FGW with a periodic neighbour kernel preserves local connectivity, whereas embeddings alone do not.
- Structure comparison across cell sizes requires an intensive representation: H/ΣZ, optimal transport, or the size-intensive FGW distance.
- MACE interaction-layer invariants are effectively element labels; for matching, the choice of layer is immaterial.
- MOF-off is chemically disjoint from OMat24; its closest OMat24 analogues are near-equilibrium AIMD frames.
- Rattled 300/500/1000 frames distort C–H bonds independently of their nominal temperature.

Next steps

- Penalise metal-to-non-metal assignments, or weight metal centres, so that C and H cannot dominate the transport cost.
- Extend the matching benchmark to several hundred materials to resolve differences between UMA, MH1 and their combination.
- Score symmetry-equivalent assignments against the ideal Alexandria parent structures.
- Obtain access to the 468 restricted files; filter frames with interatomic distances below 0.8 Å throughout.

Data, code and deliverables

Item	Location
Embedding archive (read-only)	/scratch/gpfs/ROSENGROUP/shared/aryan/saved_embeddings
Source structures (LMDB, .traj)	/scratch/gpfs/ROSENGROUP/aryan/pretrain_analysis/data
Analysis scripts	/home/sv4956/materials/analysis
Atom UMAPs; joint maps	/home/sv4956/materials/runs/atom_umap_sr1m; runs/joint_umap
MACE feature variants	/home/sv4956/materials/data/mace_variants
Matching benchmark and results	/home/sv4956/materials/data/similar_structures{,_heldout}
MOF-off vs OMat24 matrices	/home/sv4956/materials/data/structure_fgw; data/grouped_fgw
10M search, rerank results and CSVs	/home/sv4956/materials/data/mof_search
Interactive atom-matching viewer	/home/sv4956/materials/viewer

CSVs: aimd_best_matches.csv (20 MOF–OMat24 pairs with explanations) and mof_neighbours_per_subset.csv (150 pairs: five nearest neighbours per MOF in AIMD 1000 K, AIMD 3000 K and rattled relax).